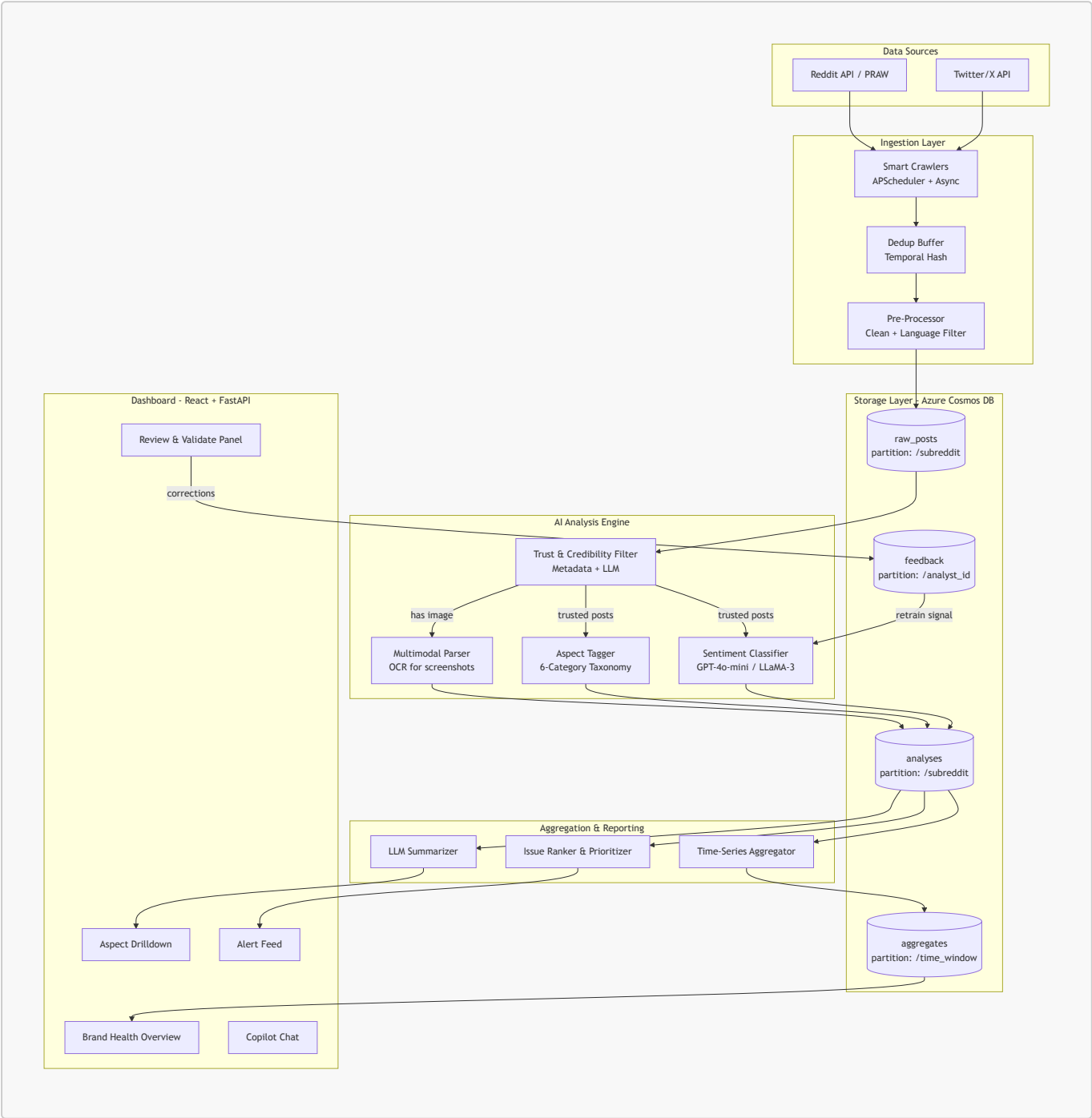
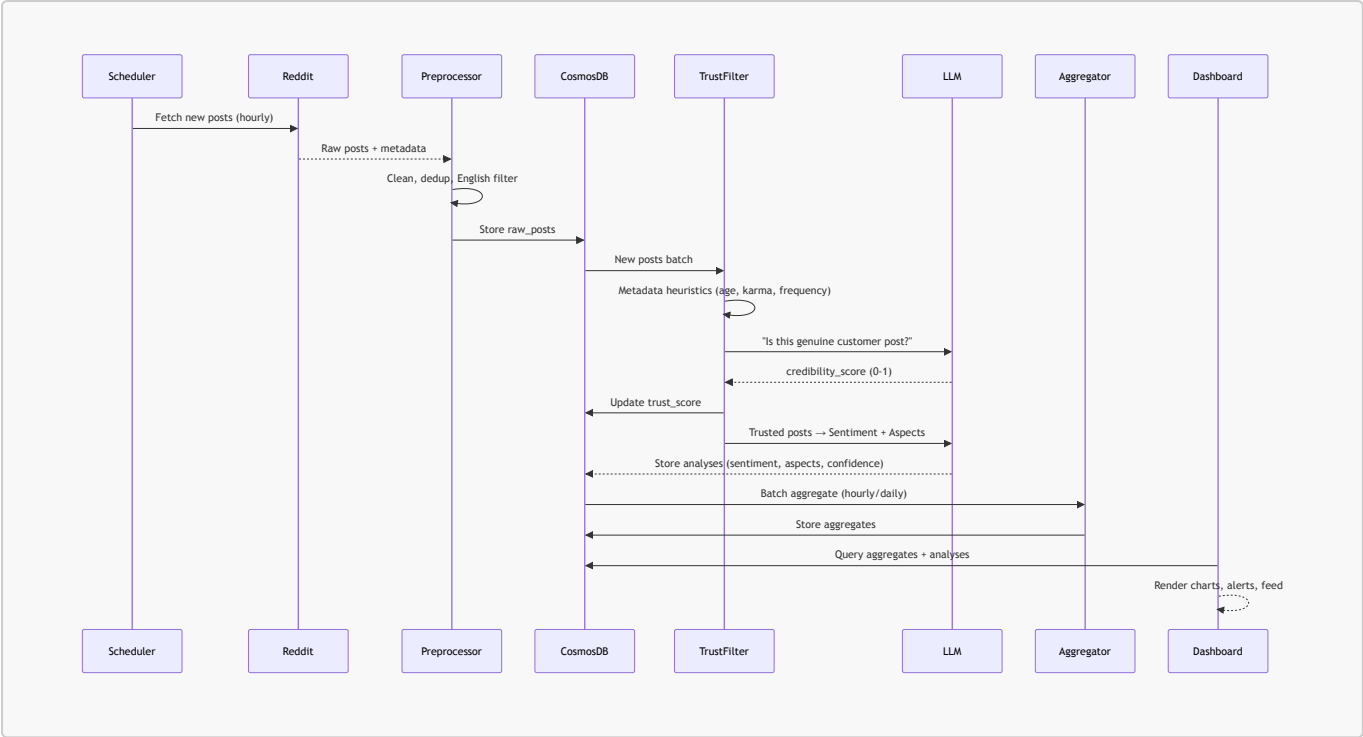


Retail Sentiment Intelligence — System Architecture

High-Level Architecture



Pipeline Flow (Step-by-Step)



Cosmos DB Data Model

Container: raw_posts

```
{
  "id": "reddit_abc123",
  "source": "reddit",
  "subreddit": "walmart",
  "author": "user_xyz",
  "title": "OGP order was completely wrong",
  "body": "Ordered 20 items, got 5 substitutions...",
  "score": 45,
  "num_comments": 23,
  "created_utc": "2026-06-01T14:30:00Z",
  "ingested_at": "2026-06-01T15:00:00Z",
  "author_metadata": {
    "account_age_days": 1200,
    "total_karma": 5600,
    "post_frequency_7d": 3
  },
  "processing_status": "analyzed",
  "partition_key": "walmart"
}
```

Container: analyses

```
{
  "id": "analysis_abc123",
  "post_id": "reddit_abc123",
  "subreddit": "walmart",
```

```

    "trust_score": 0.87,
    "trust_flags": ["genuine_account", "specific_details"],
    "sentiment": "negative",
    "sentiment_confidence": 0.92,
    "aspects": [
      {"aspect": "product_quality", "sentiment": "negative",
"confidence": 0.88},
      {"aspect": "delivery", "sentiment": "negative", "confidence": 0.75}
    ],
    "key_phrases": ["wrong items", "substitutions", "no notification"],
    "summary": "Customer received 5 incorrect substitutions in OGP order
without notification",
    "model_used": "gpt-4o-mini",
    "analyzed_at": "2026-06-01T15:05:00Z",
    "partition_key": "walmart"
  }

```

Container: **aggregates**

```

{
  "id": "agg_2026-06-01_daily",
  "time_window": "2026-06-01",
  "window_type": "daily",
  "total_posts": 342,
  "trusted_posts": 289,
  "sentiment_distribution": {
    "positive": 0.31,
    "negative": 0.48,
    "neutral": 0.21
  },
  "aspect_breakdown": {
    "delivery": {"count": 98, "avg_sentiment": -0.45},
    "product_quality": {"count": 72, "avg_sentiment": -0.32},
    "returns": {"count": 45, "avg_sentiment": -0.61},
    "customer_support": {"count": 38, "avg_sentiment": -0.55},
    "pricing": {"count": 52, "avg_sentiment": 0.12},
    "app_website": {"count": 27, "avg_sentiment": -0.38}
  },
  "top_issues": [...],
  "trending_topics": [...],
  "partition_key": "2026-06-01"
}

```

Container: **feedback** (Human-in-the-Loop)

```

{
  "id": "fb_001",
  "post_id": "reddit_abc123",
  "analyst_id": "analyst_01",

```

```
"original_sentiment": "negative",
"corrected_sentiment": "negative",
"original_aspects": ["product_quality"],
"corrected_aspects": ["product_quality", "delivery"],
"trust_override": null,
"notes": "Also involves delivery issue - wrong items delivered",
"created_at": "2026-06-02T10:00:00Z",
"partition_key": "analyst_01"
}
```

Model Recommendation

Primary: Azure OpenAI **gpt-4o-mini**

Factor	Why gpt-4o-mini
Cost	~\$0.15/1M input tokens, ~\$0.60/1M output — 60x cheaper than GPT-4o
Speed	~3x faster than GPT-4o (critical for batch processing)
Quality	82-86% on social media sentiment benchmarks with few-shot prompts
Sarcasm	Handles Reddit-style sarcasm, slang, abbreviations well
Multi-task	Single call can do sentiment + aspects + trust check
Azure	Available on Azure OpenAI (Walmart has Azure enterprise)
Structured Output	Supports JSON mode for reliable parsing

Comparison Matrix

Model	Sentiment F1	Aspect F1	Cost/1K posts	Latency	Local?
gpt-4o-mini ★	~0.84	~0.78	~\$0.08	200ms	No
gpt-4o	~0.88	~0.82	~\$2.50	600ms	No
LLaMA-3.1-8B-Instruct	~0.76	~0.70	Free (GPU)	150ms	Yes
Mistral-7B-Instruct	~0.74	~0.68	Free (GPU)	120ms	Yes
cardiffnlp/twitter-roberta	~0.72	N/A	Free	20ms	Yes
DeBERTa-v3 (fine-tuned)	~0.80	~0.74	Free	30ms	Yes

Recommended Strategy (Hybrid):

- 1. **gpt-4o-mini** — Primary analysis engine (sentiment + aspects + trust LLM check)
- 2. **cardiffnlp/twitter-roberta-base-sentiment-latest** — Fast baseline comparison + fallback
- 3. **LLaMA-3.1-8B** — Optional fine-tuning experiment for dissertation bonus points

6-Aspect Retail Taxonomy

Aspect	What it Captures	Example Posts
delivery	OGP, shipping, Spark driver, timing, missing items	"My spark driver left groceries at wrong door"
product_quality	Freshness, damage, substitutions, brand quality	"Great Value chips taste like cardboard now"
returns	Return process, refunds, exchange policy	"Tried to return online order in-store, denied"
customer_support	Associate interactions, phone/chat support, responsiveness	"Called 3 times, no one could find my order"
pricing	Price accuracy, rollbacks, competitor comparison, value	"Same item is \$3 cheaper at Target"
app_website	App crashes, website UX, Walmart+ features, checkout bugs	"App crashed during checkout, lost my cart"